

Olist E-Commerce SQL Analysis - Business Summary

Data Overview

The Olist Brazilian E-Commerce dataset contains transactional information on customers, orders, products, payments, sellers, and reviews. The dataset covers September 2016 to October 2018 and was analyzed using SQL to understand revenue trends, customer spending, purchasing behavior, and product performance. Some data-quality issues were identified, including missing product category values and unusually low revenue in the final months, which may indicate incomplete data.

Business Insights

A. Revenue grew significantly but showed monthly volatility.

Revenue increased from \$252.24 in September 2016 to over \$1.16 million in April 2018. November 2017 recorded the largest month-over-month increase of \$415,204.92, while December 2017 declined by \$316,481.32. This shows strong overall growth but also significant fluctuations in monthly performance.

B. Most customers are low- to medium-value customers.

The customer segmentation analysis found that 49.13% of customers were Medium spenders and 46.19% were Low spenders, meaning 95.32% of customers belonged to these two groups. Only 4.67% were High spenders, indicating an opportunity to increase customer spending through targeted promotions, cross-selling, and personalized offers.

C. Customer retention is a major opportunity.

The analysis found that 96.88% of customers were one-time buyers, while only 3.12% were repeat customers. This suggests that the business has a significant gap in customer retention and could potentially generate additional revenue by encouraging existing customers to purchase again.

Recommendations

A. Improve customer retention.

Introduce post-purchase campaigns, personalized recommendations, loyalty rewards, and targeted discounts to encourage one-time customers to make repeat purchases. Increasing the repeat-customer rate from its current 3.12% could significantly improve customer lifetime value.

B. Increase spending among Low and Medium customers.

Use product bundles, cross-selling, upselling, and personalized promotions to move customers toward higher spending levels. High-performing categories such as Health & Beauty, which generated \$1,258,681.34 or 9.26% of total product revenue, can be used as a focus area for targeted campaigns.